

Lesson 7 — Loop Bodies & Systematic Debugging

Matches LEGO Teacher Portal lesson: "Dizzy Descent".

Objective (I can...): I can put several actions in one loop body and debug a loop using a trace. **CS concept:** Multi-statement loop bodies; debugging discipline (reproduce → isolate → fix → retest) · **Builds on:** Lesson 6 (single-action count loop).

AI Connection (2 min): Each training round repeats the same multi-step update on new data. *First-principles question:* "If one step inside that repeated round is wrong, how many times does the mistake get made?"

Materials: LEGO CS & AI kit + device per pair, "broken program" cards, notebooks. **Model reuse:** Loops model from Lesson 6 — code only.

60-minute flow - Warm-up (8): "Find the defect" — coach shows a short block program on the board with one visible bug; pairs locate it and state what it would do wrong before any fix. Introduce the four-step debugging routine. - **Concept + demo (10):** Build a loop body with three actions (drive → tone → wait); run it, point out the whole body repeats as a unit. Then plant a bug (one block outside the loop) and model the debugging routine aloud. - **Build/Code (30):** Pairs code a 3-action repeating pattern inside one loop. **Debug task (5 min):** each pair receives a broken program (a block sits outside the loop, or the count is off); they reproduce the symptom, isolate the line, fix it, and retest, logging each step in the notebook. *Swap at 15 min.* - **Challenge (7):** Add a fourth action to the loop body and re-verify the pattern still holds together. - **Share + pack-up (5):** Pairs demo the pattern and report how they isolated their bug. Model stays built.

Pair roles: Driver codes the loop body, Navigator leads the debugging routine and logs it. *Swap at 15 min. (Trio: third owns reproducing the symptom.)*

Differentiation: *Support* — a 2-action body with a worked example. *Extension / fast-finisher* — a 5-action body plus a planted bug of their own for another pair. *Reteach* — build the body outside a loop, confirm it works, then drag it inside.

Exit check: Pair runs a loop with a 3+ action body and can describe the four-step routine they used to fix the bug. **At-home (optional):** Describe a real bug you've seen (a game glitch, a jammed printer) and how you'd isolate its cause. ~10 min.